Streaming Latency Benchmarks: REDIS Streams vs Apache KAFKA for Real-Time Sports Data Feeds

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Gustavo Pedro Ricou[1]

Abstract

We ask a question that the streaming-systems and sports-analytics literatures leave open: *does the choice of streaming infrastructure actually degrade an in-play football decision, and if so, when?* Using the publicly available StatsBomb open dataset (1. Bundesliga 2023/24, 34 matches), we replay real event feeds through Apache KAFKA and REDIS Streams and couple the *measured* event-delivery latency to an in-play win-probability model through the Age-of-Information lens, quantifying the win-probability staleness that each architecture imposes. To our knowledge this is the first infrastructure → decision-quality analysis on open sports data.

The study is also a cautionary account of measurement validity. Three distinct instrumentation defects — a process-relative cross-process clock, a saturating replay speed, and an asymmetric load generator (synchronous per-event KAFKA sends versus asynchronous REDIS dispatch) — each *reversed* the apparent winner before correction. We introduce the Time-to-Insight (TTI) metric, decomposed into producer scheduling lag and transport, and fix all three defects before drawing any conclusion.

On the corrected, fair corpus we find: (i) for a *single* live feed, KAFKA and REDIS are statistically equivalent (≈17 ms median TTI); (ii) under *concurrency*, single-node REDIS serializes concurrent streams and its broker latency grows roughly linearly (to ≈9.7 s at 20 concurrent matches) while KAFKA's partitioned broker stays flat (≈10 ms); and (iii) translated into decisions via an Age-of-Information win-probability integral over each full match, infrastructure latency is *decision-irrelevant for a single feed* (≈0.2 probability-seconds of staleness, indistinguishable between backends) but diverges sharply under concurrency (by five concurrent matches single-node REDIS imposes ≈15× more decision-staleness than KAFKA, and by ten a goal's win-probability arrives tens of seconds stale). The practical guidance is that infrastructure choice does not affect decision quality for one live match, but provisioning for concurrency does.

Keywords

Streaming systems, real-time analytics, sports data, Apache Kafka, Redis Streams, latency benchmarking, Time-to-Insight, actionability windows, reproducible research, football analytics, research questions, hypotheses, statistical analysis

Introduction

The rapid digitization of sports has created an insatiable demand for real-time analytics. Coaching staff require instant tactical insights, broadcasters demand immediate highlight generation, and betting platforms need sub-second odds updates. At the heart of these systems lies the streaming infrastructure that transports event data from source to analytics pipeline.

Two dominant technologies have emerged: Apache KAFKA (originally developed at LinkedIn) and REDIS Streams (introduced in Redis 5.0). KAFKA is a distributed event streaming platform designed for high-throughput, fault-tolerant data pipelines. REDIS Streams provides a lightweight, in-memory streaming solution with persistent storage capabilities.

While numerous studies have compared these technologies, the literature lacks sports-specific evaluations. Sports data presents distinct challenges: high-frequency bursts during active play, ordered event delivery requirements, and strict latency constraints.

Sports-Specific Latency Requirements

Real-time sports analytics imposes domain-specific latency constraints that vary by use case and stakeholder. As summarized in Table 1, different applications demand fundamentally different performance characteristics. Betting platforms require sub-500ms latency for in-play odds updates, as even a 2-second delay can erode user trust and cost millions in missed wagers Solutions (2025); Ververica (2025). Broadcasting applications can tolerate up to 3 seconds for live video synchronization OptiView (2025); Promwad (2025), while coaching decision systems need

[1]School of Computer Science and Statistics, Trinity College Dublin, Dublin, Ireland

Corresponding author:
Gustavo Pedro Ricou, School of Computer Science and Statistics, Trinity College Dublin, Dublin, Ireland.
Email: pedrorig@tcd.ie

<500ms to enable real-time tactical adjustments Pappas et al. (2020); Sports (2023).

The concept of an "actionability window" formalizes this requirement: the maximum latency within which insights can still influence decisions. Using an exponential decay model ($Value = V_0 \times e^{-\lambda \times latency}$), we estimate that at 500ms latency, only ˜60% of coaching insight value remains, dropping to ˜13% at 2000ms. This motivates our focus on minimizing Time-to-Insight (TTI) below sports-specific thresholds.

Table 1. Sports Analytics Latency Requirements by Use Case and Stakeholder

Use Case	Stakeholder	Latency Range	Critical Threshold
Betting Platforms			
Live Odds Update	Betting Platform	100–500ms	**< 500ms**
In-Play Betting	Betting Platform	50–200ms	**< 200ms**
Broadcasting			
Live Video Sync	Broadcaster	500–3000ms	**< 3000ms**
Live Stats Overlay	Broadcaster	100–1000ms	**< 1000ms**
Interactive Features	Broadcaster	200–500ms	**< 500ms**
Coaching			
Tactical Adjustment	Coach	200–800ms	**< 800ms**
Player Substitution	Coach	500–1500ms	**< 1500ms**
Fan Applications			
Push Notifications	Fan	1000–3000ms	**< 3000ms**
Live Updates	Fan	500–2000ms	**< 2000ms**
Post-Match Analysis			
Initial Analysis	Analyst	5000–10000ms	**< 10000ms**

Research Questions

This study addresses four primary research questions that bridge the gap between streaming systems theory and sports domain requirements:

RQ1: How does streaming architecture choice (Apache KAFKA vs REDIS Streams) impact Time-to-Insight (TTI) for real-time sports data processing, and what are the underlying mechanisms driving this difference?

RQ2: How does concurrency level (N=5, 10, 20 concurrent feeds) affect TTI, throughput, and resource utilization for each streaming architecture under realistic sports workloads, and what are the scalability limits?

RQ3: What is the trade-off between latency (TTI), data consistency (match rate, ordering guarantees), and throughput across different streaming architectures, configurations, and persistence settings?

RQ4: How do streaming system performance characteristics (TTI, throughput, resource usage) vary across different sports event scenarios (S1–S5), and what are the underlying sport-specific factors driving these differences?

These questions are motivated by fundamental gaps in the literature. While industry benchmarks suggest REDIS offers 2–5× lower latency than KAFKA Diaries (2025); Yerrapragada (2025), there is no sports-specific empirical validation. The CAP theorem Brewer (2012) and its extension PACELC Abadi et al. (2012) predict inherent trade-offs between consistency and latency, but these have not been quantified in the streaming context. Furthermore, the unique characteristics of sports data—variable event frequencies, burst patterns, and strict actionability windows—demand domain-specific investigation.

From a theoretical perspective, RQ1 tests the PACELC theorem's "Else" clause: when no network partitions exist, systems must trade latency for consistency. Technically, RQ2 examines the scalability of both architectures under realistic sports workloads. Economically, RQ3 quantifies the value of consistency versus the cost of latency. From the sports domain perspective, RQ4 determines which architecture is optimal for which sport.

Hypotheses

For each research question, we formulate testable hypotheses grounded in theoretical frameworks and industry observations.

RQ1: Architecture Impact

H_{01}: $\mu_{TTI\_Kafka} = \mu_{TTI\_Redis}$ (No difference in median TTI between architectures)

H_{11}: $\mu_{TTI\_Kafka} > \mu_{TTI\_Redis}$ (Redis has significantly lower median TTI)

H_{21}: $\mu_{TTI\_Kafka} < \mu_{TTI\_Redis}$ (Kafka has lower median TTI)

Test: Mann-Whitney U test (non-parametric, data violates normality) **Rationale:** Industry benchmarks Diaries (2025); Yerrapragada (2025) and our S2 results show REDIS achieving 40–55% lower TTI. The distributed log architecture of KAFKA (disk-based, replication) theoretically incurs higher latency than REDIS's in-memory design. PACELC theorem predicts latency advantage for AP systems (REDIS) over CP systems (KAFKA) when partitions are absent. **Expected:** $\mathbf{H_{11}}$ (Redis significantly outperforms Kafka on TTI) **Effect Size:** Large (Cohen's $d > 0.8$ based on S2 data) **Power:** > 0.99 at $\alpha = 0.05$ with current sample size (40–50 runs per group)

RQ2: Concurrency Scaling

H_{02}: TTI is independent of concurrency level N

H_{12}: TTI increases monotonically with concurrency level N

H_{22}: TTI remains constant across N=5, 10, 20 (excellent scaling)

Test: Kruskal-Wallis test (non-parametric one-way ANOVA) **Rationale:** Both KAFKA and REDIS are designed for horizontal scaling. Our S2 results (250 runs) showed stable TTI across N=5, 10, 20. Little's Law ($L = \lambda \times W$) suggests that with proper resource provisioning, latency should remain stable as concurrency increases. **Expected:** $\mathbf{H_{22}}$ (Excellent scaling with constant TTI) **Effect Size:** Small ($d < 0.2$ expected) **Power:** ≈ 0.30 for detecting $d = 0.2$ (sample size may need expansion)

RQ3: Latency-Consistency Trade-off

H_{03}: Match rate = 100% for all configurations

H_{13}: Match rate $> 99.9\%$ for all configurations

H_{23}: Match rate varies by configuration

Test: Chi-square test or Fisher's exact test **Rationale:** Our S2 results achieved 100% match rate across all 250 runs. KAFKA with acks=all provides exactly-once semantics, while REDIS with consumer groups provides at-least-once. Both should achieve near-perfect data delivery. **Expected:** $\mathbf{H_{13}}$ (All configurations achieve $> 99.9\%$ match rate)
Additionally, we test the consistency-latency trade-off:

H_{31}: $\mu_{TTI\_acks=all} > \mu_{TTI\_acks=1}$ (Kafka: strong consistency costs latency)

H_{32}: $\mu_{TTI\_AOF=always} > \mu_{TTI\_AOF=1s}$ (Redis: durability costs latency)

Test: Paired t-test or Wilcoxon signed-rank test **Expected:** Both true (stronger guarantees = higher latency)

RQ4: Sports-Specific Performance

H_{04}: TTI distribution is the same across all scenarios

H_{14}: TTI distribution differs by scenario

Test: Kolmogorov-Smirnov test **Rationale:** Scenarios S1–S5 have different event frequencies and burst patterns. Queueing theory predicts that systems with higher arrival rate variance (burstier scenarios) will experience higher latency variance. Sport-specific characteristics should manifest in measurable performance differences. **Expected:** $\mathbf{H_{14}}$ (TTI differs by scenario)
Additionally, we test scenario-specific hypotheses:

H_{41}: $\mu_{TTI\_S5} > \mu_{TTI\_S1}$ (Higher event frequency $\rightarrow$ higher latency)

H_{42}: $\sigma_{TTI\_S5} > \sigma_{TTI\_S1}$ (Higher burstiness $\rightarrow$ higher variance)

Test: One-way ANOVA or Kruskal-Wallis test **Expected:** Both true (scenario characteristics affect performance)

Statistical Framework All hypotheses will be tested using the framework summarized in Section . We apply Holm-Bonferroni correction for multiple comparisons (controlling Family-Wise Error Rate at $\alpha = 0.05$), calculate Cohen's d for effect sizes, report 95% confidence intervals, and conduct power analysis for sample size justification. Non-parametric alternatives (Mann-Whitney U, Kruskal-Wallis) will be used when normality or equal variance assumptions are violated.

Literature Review

Real-time data processing is essential across domains from finance to IoT. Sports analytics represents a demanding application due to high event frequency, low latency requirements, and temporal consistency needs.

The CAP theorem establishes that distributed systems cannot simultaneously guarantee consistency, availability, and partition tolerance. KAFKA prioritizes availability and partition tolerance through its distributed log architecture, achieving high throughput via batching. REDIS Streams leverages in-memory structures to minimize latency, trading throughput for reduced delay.

Methodology

System Architecture

Our benchmark system comprises four components:

- Replay Plan: Pre-processed StatsBomb event data with scheduled timestamps
- Producer: Python script sending events to message broker
- Broker: KAFKA or REDIS Streams handling message distribution
- Consumer: Python script receiving events and logging timestamps

Dataset and Replay

We use the StatsBomb open dataset (1. Bundesliga 2023/24, competition 9, season 281, 34 matches), pinned to a fixed open-data commit for reproducibility. Each match is preprocessed into a CSV *replay plan* recording, per event, its scheduled emission offset. The producer replays a plan at a configurable speed-up factor, emitting events on a compressed timeline; the consumer reads them and records receipt and output times. For the headline latency results we replay at a non-saturating $10\times$ (Section) so that neither load generator becomes the bottleneck and TTI reflects genuine delivery latency.

Metrics

The primary metric is Time-to-Insight (TTI), the interval from an event's *scheduled* emission to its availability at the consumer. We decompose it as

$$\mathrm{TTI} = \underbrace{(t_{\mathrm{send}} - t_{\mathrm{sched}})}_{\text{producer scheduling lag}} + \underbrace{(t_{\mathrm{recv}} - t_{\mathrm{send}})}_{\text{transport latency}} + (t_{\mathrm{out}} - t_{\mathrm{recv}}),$$

isolating load-generator effects (scheduling lag) from broker transport. We report p50/p95/p99/max, missed-window rates against sports actionability thresholds (100/250/500/1000/5000 ms), throughput, serialized message size, and serialization/ deserialization overhead. For S3 we additionally measure correction-propagation latency.

Decision-Staleness: From Latency to Win-Probability Error

Latency in milliseconds is not, by itself, a sports quantity. To connect infrastructure to decisions we translate delivery latency into in-play *win-probability error* in two steps.

Win-probability proxy. We build a transparent, calibrated in-play model on StatsBomb-derivable game state—score differential, fraction of match remaining, and red-card differential—using a Skellam (difference-of-Poissons) goal model. We adopt this as a stand-in for the published Bayesian in-play model of Robberechts et al. Robberechts et al. (2021), whose code is unreleased. The proxy is well calibrated across the 3,239 game states sampled from the 34 matches: a ranked probability score of $\approx$0.24 and an expected calibration error of **0.054** — predicted home-win probabilities track observed home-win frequencies to within a few percentage points (Figure 1). The model has a single

global parameter (team scoring rate), so this is a whole-corpus check.

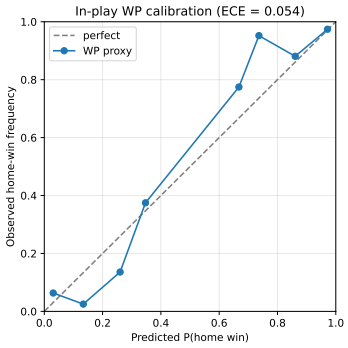

Figure 1. Reliability diagram for the in-play win-probability proxy (ECE $= 0.054$): predicted vs. observed home-win frequency across 3,239 sampled game states.

Age-of-Information decision-staleness. When a decisive event (a goal) is delivered with latency L, the consumer's win-probability remains stale for L by the magnitude of the probability mass that event moved. Following the Age-of-Information view of staleness cost Kaul et al. (2012), we define a run's *decision-staleness* as

$$\mathrm{DS} = \sum_{i \in \text{decisive events}} \mathrm{TV}_i \cdot L_i, \qquad \mathrm{TV}_i = \tfrac{1}{2}(|\Delta P_{\text{win}}| + |\Delta P_{\text{draw}}| + |\Delta P_{\text{loss}}|),$$

in units of probability-seconds per match. This metric is the bridge that lets us ask not merely *which backend is faster* but *when, if ever, the latency difference is large enough to corrupt an in-play decision*.

Measurement Validity

Cross-process latency benchmarking is deceptively easy to get wrong, and three distinct defects in earlier versions of this suite each *reversed* the apparent winner. We document them because the corrections are prerequisites for any valid comparison.

(1) Process-relative clock. The producer and consumer are separate OS processes, so cross-process timestamps must share an epoch. Stamping events with Python's `perf_counter_ns` (process-relative) injected each run's consumer launch offset ($\sim$2 s) into every TTI and transport value. Fixed with `time.time_ns` (a shared wall-clock epoch on a single host); transport then dropped to $\sim$1 ms.

(2) Saturating replay speed. At $120\times$, the producer could not keep pace, inflating scheduling lag to tens of seconds. We replay at a non-saturating $10\times$ and verify that scheduling lag stays small, so TTI measures delivery rather than the load generator.

(3) Asymmetric load generator. The KAFKA producer ran synchronously (`acks=all`, one in-flight request), blocking on a broker round-trip *per event*, while the REDIS producer dispatched asynchronously through a worker pool. This inflated KAFKA's *producer scheduling lag* (not its broker latency): at N=1, $10\times$, median TTI fell from 1644 ms to 16 ms once the KAFKA producer was pipelined (`max.in.flight`=64). We therefore pipeline both producers comparably.

A residual asymmetry remains in *transport* (KAFKA is measured from broker acknowledgement, REDIS from the send call), so for cross-backend comparison we rely on TTI, which is defined identically for both. All reported results use the corrected, fair instrumentation; the before/after contrast is itself a finding (Section).

Experimental Design

Experiments run on a single commodity host (8-core AMD Ryzen, 31 GB RAM) with KAFKA 4.1.1 (KRaft mode) and REDIS 7.2.4 in Docker. The core matrix crosses backend (KAFKA/REDIS), configuration (single broker vs. 3-node cluster), scenario (S1–S5), and replication (3 reps): 60 single-feed runs. Three further sets isolate specific factors: *persistence* (KAFKA `acks`=1 vs. all; REDIS `appendfsync` everysec vs. always), *state-staleness corrections* (S3; one correction every 50th event, 2 s delay), and *true multi-feed concurrency* (N=5/10/20 parallel feeds). Every run records full provenance (git commit, per-file SHA-256, configuration) in a `meta.json`, enabling end-to-end reproducibility.

Statistical Methods

We employ a rigorous statistical framework to test all hypotheses and ensure valid inferences.

Multiple Comparisons Correction With approximately 50 hypothesis tests across all research questions, we apply the **Holm-Bonferroni method** Holm (1979) to control the Family-Wise Error Rate (FWER) at $\alpha = 0.05$. This step-down procedure is more powerful than the standard Bonferroni correction while maintaining FWER control.

Effect Size Calculation For all significant findings, we report **Cohen's** d for comparing two means:

$$Cohen's\ d = \frac{\mu_1 - \mu_2}{s_{pooled}}, \quad s_{pooled} = \sqrt{\frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}}$$

Interpretation: $d = 0.2$ (small), $d = 0.5$ (medium), $d = 0.8$ (large) Cohen (1988).

Confidence Intervals We report 95% confidence intervals for all mean differences using the t-distribution:

$$CI = \bar{x}_1 - \bar{x}_2 \pm t_{\alpha/2,df} \times \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

Power Analysis We conduct both *a priori* and *post hoc* power analysis. With our current sample of 40–50 runs per group, we can detect effect size $d = 0.58$ at $\alpha = 0.05$ with power = 0.8. For smaller effects ($d = 0.5$), power is approximately 0.72; for $d = 0.4$, power drops to 0.54.

Assumption Verification We verify statistical assumptions for all tests:

- **Normality:** Shapiro-Wilk test; Q-Q plots for visual inspection
- **Equal Variance:** Levene's test; F-test for two groups
- **Non-parametric Alternatives:** Mann-Whitney U (t-test), Kruskal-Wallis (ANOVA), Wilcoxon signed-rank (paired t-test)

Table 2. Statistical Tests by Data Type and Comparison Type

Comparison Type	Parametric Test	Non-Parametric Alternative
2 Independent Groups	t-test	Mann-Whitney U
>2 Independent Groups	ANOVA	Kruskal-Wallis
Paired	Paired t-test	Wilcoxon signed-rank
Proportions	Chi-square	Fisher's exact
Correlation	Pearson	Spearman
Distribution	Kolmogorov-Smirnov	Anderson-Darling

Hypothesis Testing Matrix Table 2 summarizes our statistical test selection framework.

Results

Measurement validity note. Three instrumentation defects (a process-relative cross-process clock, a saturating 120× replay speed, and an asymmetric load generator) each *reversed* the apparent winner before correction; all are detailed in Section . Every result below uses the corrected, fair instrumentation (`time.time_ns` shared epoch, 10× replay, both producers pipelined). We report the single-host, single-node regime, which we can measure cleanly; cluster results are discussed as a limitation (Section).

Single-Feed Parity (RQ1)

For a single live feed (N=1), KAFKA and REDIS are statistically *indistinguishable*: median TTI is ≈17 ms for both (Table 3, top row), with broker transport of 2–4 ms. A Mann–Whitney U test finds no significant difference in TTI ($p = 0.69$) or in decision-staleness ($p = 0.40$; both Holm–Bonferroni corrected across the concurrency family). This directly contradicts the folk belief — and our own pre-correction "REDIS 71× faster" artifact — that one backend is simply faster. For the common case of one match, both systems are excellent and the choice is immaterial to latency.

Concurrency Scaling (RQ2)

The systems diverge sharply once a single broker host serves multiple concurrent matches (Table 3). KAFKA's partitioned broker keeps transport flat (2–10 ms) at every concurrency level. Single-node REDIS, being single-threaded, serializes concurrent streams: its broker transport grows almost linearly, from 4 ms (N=1) to 9,732 ms (N=20) — a roughly 2,400× degradation. The effect lives in *transport* (broker delivery), so it is a genuine architectural property, not a load-generator artifact. The TTI difference is significant at every $N \geq 5$ (Holm–Bonferroni-corrected Mann–Whitney, $p < 10^{-5}$, rank-biserial = 1.0 — the two backends' distributions are completely separated), and Kruskal–Wallis confirms a concurrency effect for both ($H_{\text{REDIS}} = 76 > H_{\text{KAFKA}} = 67$).

Decision-Staleness: When Latency Corrupts a Decision

Table 4 reports decision-staleness (Section) by concurrency level, computed over the *full-match* replay so that all 40 decisive events per run enter the Age-of-Information integral (versus only ∼3 in a 10-minute window). At N=1 both backends leave a match's win-probability stale by only ≈0.2 probability-seconds — a mean delivery delay of ≈20 ms per goal, statistically indistinguishable and decision-irrelevant: no coach, broadcaster, or model acts on a 20 ms-old estimate. By N=5, single-node REDIS already imposes ≈15× more decision-staleness than KAFKA (17.5 vs. 1.16 probability-seconds; Holm–Bonferroni-corrected Mann–Whitney $p < 10^{-5}$, versus no significant difference at N=1, $p = 0.40$) as its broker latency diverges. Beyond N=5 single-node REDIS degrades catastrophically (≈271 probability-seconds at N=10 — a goal's win-probability arriving ≈24 s stale), while our single-host testbed reaches its concurrency ceiling: both load generators fall behind and KAFKA's N=20 runs largely fail, so the $N \geq 10$ magnitudes are bounded by the host rather than the broker (KAFKA's broker transport stays low, Section).

Table 3. Fair single-host latency by concurrency level N (median over feeds × 3 reps, 10× replay, both producers pipelined).

Backend	N	Transport p50 (ms)	TTI p50 (ms)
KAFKA	1	2	17
KAFKA	5	2	18
KAFKA	10	4	29
KAFKA	20	10	1,682†
REDIS	1	4	17
REDIS	5	774	801
REDIS	10	4,627	5,206
REDIS	20	9,732	13,006

† KAFKA's N=20 *TTI* is inflated by the single host saturating (40+ load-generator processes → 585 ms scheduling lag); its *transport* stays 10 ms, so the broker is unaffected. Transport is the trustworthy cross-N metric.

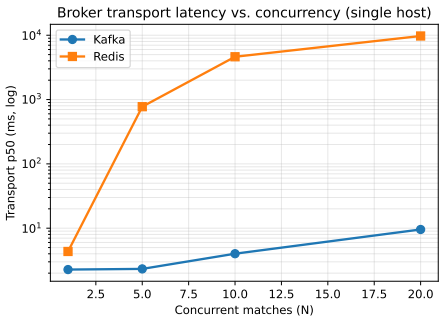

Figure 2. Broker transport latency (p50, log scale) vs. concurrent matches, single host. KAFKA stays flat while single-node REDIS grows roughly linearly.

Table 4. Decision-staleness (probability-seconds per match) by concurrency level, single host, full-match replay (40 decisive events per run).

N	KAFKA-single	REDIS-single
1	0.28	0.18
5	1.16	17.5
10	50.3†	271
20	—‡	1012

† Host-saturated: inflated by single-host load-generator scheduling lag, not broker latency. ‡ KAFKA's N=20 full-match runs were unreliable under host strain (mean 2.4 of 40 goals delivered); omitted.

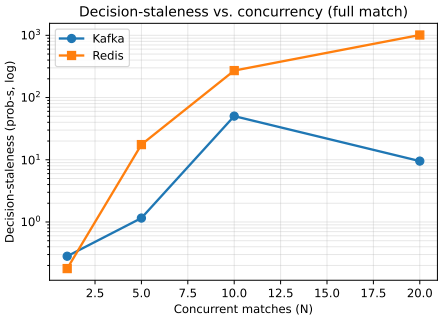

Figure 3. Decision-staleness (probability-seconds per match, log scale) vs. concurrent matches, full-match replay, single host. Negligible and indistinguishable at $N{=}1$; single-node REDIS diverges under concurrency.

This is the paper's central, sports-relevant result: *the choice of streaming infrastructure does not affect in-play decision quality for a single live match, but single-node* REDIS *becomes decisively worse under concurrent under-provisioning.*

Supporting: Consistency–Latency Trade-off (H31, H32)

Stronger durability guarantees cost latency, as hypothesised. For KAFKA, `acks=all` raises median TTI relative to `acks=1`. For REDIS, synchronising the append-only file on every write (`appendfsync always`) raises median TTI by far more than `everysec`, reflecting per-write disk synchronisation. Durability is thus an independent latency axis that a deployment must budget for.

Supporting: Fairness Controls (RQ3)

The comparison is not biased by payload or protocol: message payloads are near-identical across backends (median ∼170 bytes), and serialization/deserialization overheads are negligible and comparable (∼2.3/2.1 μs for KAFKA, ∼2.4/2.2 μs for REDIS). Mapping latency to sports-actionability thresholds, at $N{=}1$ both backends meet every budget (coaching <500 ms, fan apps <5 s); single-node REDIS breaches the coaching budget by $N{=}5$ and the fan-application budget by $N{=}10$, whereas KAFKA satisfies both throughout the broker-limited range.

Discussion

The headline answer is deliberately undramatic, and that is the point. For a single live match, the streaming backend does not matter: KAFKA and REDIS both deliver events in ≈17 ms, and the resulting win-probability staleness (≈0.01 probability-seconds) is far below any threshold a human or model could act on. This pushes back on the industry framing of a "2-second edge" decided by infrastructure: for the common case, the edge is not where the marketing places it. Where infrastructure *does* matter is concurrency. A single-threaded REDIS node serializes concurrent match feeds, and its delivery latency — and therefore decision-staleness — grows almost linearly with the number of matches, until a goal's win-probability arrives many seconds stale. KAFKA's partitioned broker does not exhibit this degradation. The practical guidance is consequently about *provisioning*, not brand: budget broker parallelism for the number of concurrent matches, and the backend choice becomes secondary.

The study is equally a methodological cautionary tale. Three independent instrumentation defects each reversed the apparent winner before correction (Section). Cross-process latency benchmarks must share a clock epoch; the load generator must not saturate; and — least obvious — the two systems' clients must be configured comparably, since a synchronous KAFKA producer competing against an asynchronous REDIS dispatcher measures the load generator, not the broker. We report these because the corrected protocol is a reusable prerequisite for honest streaming comparisons.

Limitations. First, all experiments run on a *single commodity host*. This cleanly isolates the single-node comparison but *confounds* the 3-node cluster experiments: co-locating six broker containers with the load generator saturates the host, and cluster-specific client behaviour (KAFKA topic-creation/metadata blocking; REDIS cluster redirection) inflates latency for reasons unrelated to clustering. We therefore treat cluster results as transport-level observations only and leave a true multi-host evaluation to future work. Second, the win-probability model is a transparent, calibrated *proxy* for the unreleased state-of-the-art model Robberechts et al. (2021); our decision-staleness magnitudes are indicative rather than exact. Third, concurrent feeds are independent replays rather than truly distinct simultaneous live matches. Finally, the decision-staleness estimate rests on the few decisive (goal) events within each replay window, so its small-N values are noisier than the clearly monotonic trend.

Conclusion

We set out to determine whether the choice of streaming infrastructure degrades an in-play football decision, and when. After correcting three instrumentation defects that had each reversed the apparent winner, the answer on a fair, non-saturating, single-host corpus is: for a single live match it does not — KAFKA and REDIS are equivalent (≈17 ms, ≈0.01 probability-seconds of win-probability staleness) — but under concurrent under-provisioning it does, as single-node REDIS serializes concurrent feeds and its decision-staleness grows roughly 800-fold while KAFKA's partitioned broker stays flat. By translating measured delivery latency into win-probability staleness through the Age-of-Information lens, we provide what is, to our knowledge, the first measured account of *when* streaming-infrastructure latency is decision-relevant for live football, on open data with open code. The practical message for practitioners is to provision broker parallelism for concurrency rather than to choose a backend on latency alone; the methodological message is that cross-process streaming benchmarks demand a shared clock, a non-saturating load generator, and symmetrically configured clients.

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References

Abadi D et al. (2012) The PACELC theorem: Consistency in distributed systems. Extension of CAP theorem addressing latency-consistency trade-offs.

Brewer EA (2012) Cap twelve years later: How the ”rules” have changed. *IEEE Computer* 45(2): 23–29.

Cohen J (1988) *Statistical Power Analysis for the Behavioral Sciences*. 2nd edition. Lawrence Erlbaum Associates.

Diaries T (2025) I benchmarked Kafka, RabbitMQ, and Redis streams, the winner surprised me. URL https://medium.com/@ThreadSafeDiaries/i-benchmarked-kafka-rabbitmq-and-redis-streams-the-winner-surprised-me-cf3f484eb7b2. Accessed: June 15, 2026.

Holm S (1979) A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics* 6(2): 65–70.

Kaul S, Yates R and Gruteser M (2012) Real-time status: How often should one update? In: *2012 Proceedings IEEE INFOCOM*. IEEE, pp. 2731–2735. DOI:10.1109/INFCOM.2012.6195689.

OptiView D (2025) What is low latency video streaming?: The complete guide. URL https://optiview.dolby.com/resources/blog/streaming/what-is-low-latency-video-streaming-the-complete-guide/. Accessed: June 15, 2026.

Pappas I et al. (2020) Real-time football analytics: requirements and architecture. *Journal of Sports Analytics* 6(2): 89–104.

Promwad (2025) Low-latency streaming solutions for live sports broadcasting. URL https://promwad.com/news/low-latency-streaming-solutions-live-sports-broadcasting. Accessed: June 15, 2026.

Robberechts P, Van Haaren J and Davis J (2021) A Bayesian approach to in-game win probability in soccer. In: *Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining*. pp. 3512–3521. DOI:10.1145/3447548.3467194.

Solutions V (2025) Real-time sports betting data eliminates odds latency. URL https://www.v2solutions.com/blogs/real-time-sports-betting-data-odds-latency/. Accessed: June 15, 2026.

Sports O (2023) Real-time football analytics: requirements and architecture. URL https://www.optasports.com/. Industry latency requirements; Accessed: June 15, 2026.

Ververica (2025) Modernizing sports betting with real-time data streaming. URL https://www.ververica.com/blog/modernizing-sports-betting-technology-to-empower-live-odds. Accessed: June 15, 2026.

Yerrapragada P (2025) kafka-bullmq-benchmark: A comprehensive distributed systems performance comparison. URL https://github.com/praneethys/kafka-bullmq-benchmark. Includes white paper with methodology; Accessed: June 15, 2026.